Introduction to Data Science

Final-term Project

The final-term project is designed to evaluate your ability to apply data science concepts across all phases of a data-driven study, with a strong focus on working with real-world data, developing a research-driven approach, and building suitable models. Unlike the mid-term project, this project requires you to collect your own dataset through web scraping and carry out the complete analysis pipeline based on a clearly defined research objective.

1. Research Objective Definition

You must begin by clearly defining a research problem or objective. This objective should clearly state what you aim to analyze or predict and should be specific, measurable, and data-driven (such as prediction, classification, trend analysis, sentiment analysis, or recommendation). Your chosen objective should naturally justify the type of data you collect and the modeling techniques you apply.

2. Data Collection (Web Scraping)

You are required to collect data using web scraping techniques from publicly available sources.

- The data source or sources must be clearly mentioned and properly justified.
- You should briefly explain the scraping methodology, tools, and libraries used (for example, R-based scraping tools).

3. Data Understanding and Exploration

After collecting the data, you are expected to develop a clear understanding of the dataset by:

- Describing the structure of the dataset, including variables, data types, and overall size.
- Performing exploratory data analysis (EDA) using suitable visualizations and summary statistics.
- Identifying missing values, outliers, noise, or any inconsistencies present in the data.

4. Data Preprocessing

Based on your research objective, you must carry out the required data preprocessing steps. These may include:

- Cleaning the data and handling missing values.
- Feature engineering and data transformation.
- Feature selection, where applicable.
- Encoding categorical variables.
- Applying normalization or scaling techniques.
- All preprocessing steps should be clearly explained and logically justified.

5. Modeling and Analysis

You must select and implement one or more appropriate models that align with your research objective.

- The choice of model or models should be theoretically justified.
- The processes of model training, validation, and testing must be clearly described.
- Appropriate performance metrics should be used, reported, and interpreted.

6. Results and Interpretation

You should:

- Present the results of your model clearly using tables, graphs, and evaluation metrics.
- Interpret the results in relation to your research objective.
- Discuss whether the objective was achieved and provide possible reasons for the outcomes.

7. Conclusion and Future Work

The project should conclude with:

- A concise summary of the key findings.
- A discussion of the limitations of the study.
- Suggestions for possible improvements or future extensions of the work.

8. Project Deliverables

You must submit:

- A well-structured and properly documented written project report.
- Source code used for web scraping, data preprocessing, and modeling.
- The final dataset or clear instructions/scripts to regenerate the dataset through web scraping.

Submission Deadline and Viva Requirement

- The final term project must be submitted on or before 4th May 2026.
- Students must carry a printed hard copy of the project report during the viva examination scheduled on 5th May 2026.